Practice workbook · Finding the first term and common ratio of a geometric series

Writing given terms as ar^(n-1), dividing to remove a, and solving for r and a, or finding a term from the nth-term formula.

17 question parts: 1 worked-example retry and 16 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-08

1. Question 4 · part a.ii · 5 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part a.ii only.

4 A geometric series G has first term a and common ratio r

The third term of G is 10 and the seventh term of G is 40

Given that the second term of G is negative,

(a) find

(i) the exact value of r

(ii) the value of a

(5)

(b) Explain why G does not have a sum to infinity.

(1)

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1. Question 4 · part a.ii · 5 marks · Working and answer

Working and answer

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2. Question 3 · part a.i · 0 marks

2022 JAN · Paper 01

Further practice · Attempt part a.i only.

3 A geometric series G has first term a and common ratio r

The 2nd term of G is $\frac{5}{16}$ and the 5th term of G is $\frac{135}{1024}$

(a) Find

(i) the value of r

(ii) the value of a

(5)

Given that G is convergent with sum to infinity S

(b) find the exact value of S

(2)

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2. Question 3 · part a.i · 0 marks · Working and answer

Working and answer

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3. Question 3 · part a.ii · 5 marks

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Further practice · Attempt part a.ii only.

3 A geometric series G has first term a and common ratio r

The 2nd term of G is $\frac{5}{16}$ and the 5th term of G is $\frac{135}{1024}$

(a) Find

(i) the value of r

(ii) the value of a **(5)**

Given that G is convergent with sum to infinity S

(b) find the exact value of S **(2)**

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3. Question 3 · part a.ii · 5 marks · Working and answer

Working and answer

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4. Question 7 · part a.i · 0 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

7 The nth term of a geometric series G is u_n

The first term of G is a and the common ratio of G is r, where $r > 0$

Given that $u_3 = 4$ and that $u_7 = 16$

(a) (i) show that $r = \sqrt{2}$

(ii) find the value of a.

(3)

(b) Find the least value of n for which $u_n > 500$

(4)

The sum of the first n terms of G is S_n

(c) Find S_{20}

Give your answer in the form $p\left(1+\sqrt{2}\right)$ where p is an integer.

(4)

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4. Question 7 · part a.i · 0 marks · Working and answer

Working and answer

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5. Question 7 · part a.ii · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

7 The nth term of a geometric series G is u_n

The first term of G is a and the common ratio of G is r, where $r > 0$

Given that $u_3 = 4$ and that $u_7 = 16$

(a) (i) show that $r = \sqrt{2}$

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(b) Find the least value of n for which $u_n > 500$

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The sum of the first n terms of G is S_n

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Give your answer in the form $p\left(1+\sqrt{2}\right)$ where p is an integer.

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5. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

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6. Question 5 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

5 A geometric series G has first term 12 and common ratio $\frac{3}{8}$

(a) Find the sum to infinity of G

(2)

(b) Show that the 6th term of G can be written as $\frac{3^6}{2^{13}}$

(3)

The nth term of G is u_n

(c) By finding an expression for u_n in terms of n, show that

$$\log_2 u_n = n\log_2 3 - 3n + 5$$

(5)

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6. Question 5 · part b · 3 marks · Working and answer

Working and answer

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7. Question 4 · part a.i · 0 marks

2025 JUN · Paper 02

Further practice · Attempt part a.i only.

4 A geometric series G has first term a and common ratio r

The third term of G is 10 and the seventh term of G is 40

Given that the second term of G is negative,

(a) find

(i) the exact value of r

(ii) the value of a

(5)

(b) Explain why G does not have a sum to infinity.

(1)

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7. Question 4 · part a.i · 0 marks · Working and answer

Working and answer

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8. Question 7 · part a · 3 marks

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Further practice · Attempt part a only.

7 The geometric series G has first term a, common ratio r and nth term u_n

Given that $u_4 = e^{x+2}$ and that $u_7 = e^{\frac{2x+1}{2}}$

(a) show that $r = e^{-\frac{1}{2}}$

(3)

(b) Hence find a in terms of e and x.

(3)

Given that the sum to infinity of G can be written as $\frac{e^p}{e^{\frac{1}{2}} - 1}$

(c) find an expression for p in terms of x.

(3)

Given that $u_{18} > 1.6$ and that x is an integer,

(d) find the least value of x.

(4)

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8. Question 7 · part a · 3 marks · Working and answer

Working and answer

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9. Question 7 · part b · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

7 The geometric series G has first term a, common ratio r and nth term u_n

Given that $u_4 = e^{x+2}$ and that $u_7 = e^{\frac{2x+1}{2}}$

(a) show that $r = e^{-\frac{1}{2}}$

(3)

(b) Hence find a in terms of e and x.

(3)

Given that the sum to infinity of G can be written as $\dfrac{e^p}{e^{\frac{1}{2}} - 1}$

(c) find an expression for p in terms of x.

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Given that $u_{18} > 1.6$ and that x is an integer,

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9. Question 7 · part b · 3 marks · Working and answer

Working and answer

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10. Question 3 · part a.i · 0 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

3 A geometric series has first term a and common ratio r, where $r > 0$

Given that the 3rd term of the series is 5 and that the 5th term of the series is $\frac{5}{2}$

(a) find

(i) the exact value of r

(ii) the value of a

(4)

(b) Find the sum to infinity of this series.
Give your answer in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

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10. Question 3 · part a.i · 0 marks · Working and answer

Working and answer

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11. Question 3 · part a.ii · 4 marks

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Further practice · Attempt part a.ii only.

3 A geometric series has first term a and common ratio r, where $r > 0$

Given that the 3rd term of the series is 5 and that the 5th term of the series is $\frac{5}{2}$

(a) find

(i) the exact value of r

(ii) the value of a

(4)

(b) Find the sum to infinity of this series.
Give your answer in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

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11. Question 3 · part a.ii · 4 marks · Working and answer

Working and answer

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12. Question 7 · part a · 3 marks

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Further practice · Attempt part a only.

7 The 7th term of a geometric series is 192 and the 8th term of this geometric series is 1152

(a) Find, as a fraction in its simplest form, the 4th term of this geometric series.

(3)

A different geometric series G has a common ratio r and nth term t_n

Given that $t_3 = 24$ and $t_2 + t_3 + t_4 = -36$

(b) show that r satisfies the equation

$$2r^2 + 5r + 2 = 0$$

(5)

Given further that G is convergent with sum to infinity S,

(c) find the value of S.

(4)

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12. Question 7 · part a · 3 marks · Working and answer

Working and answer

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13. Question 8 · part a · 1 mark

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

8 The nth term of a geometric series G is U_n and the sum of the first n terms of G is S_n

Given that $U_n = \frac{25}{4}\left(\frac{3}{5}\right)^n$

(a) find the exact value of U_5

(1)

(b) Show that $S_n = \sum_{r=1}^{n} \frac{A}{B}\left(\frac{3}{5}\right)^{r-1}$ where A and B are integers to be found.

(3)

The sum to infinity of G is S

(c) Find the least value of n such that $S - S_n < 0.045$

(6)

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13. Question 8 · part a · 1 mark · Working and answer

Working and answer

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14. Question 6 · part a · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

6 (i) A geometric series G has first term $\left(3+\sqrt{3}\right)$ and common ratio r

The second term of G is $\sqrt{3}$

(a) Find, showing all your working, the exact value of r

Give your answer in the form $\frac{p+\sqrt{3}}{q}$ where p and q are integers to be found.

(2)

(b) Explain why G is convergent.

(1)

(ii) A different geometric series H has first term 8 and common ratio 0.6

The sum to n terms of H is S_n and the sum to infinity of H is T

Find, using logarithms, the least value of n such that $T - S_n < 0.12$

(5)

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14. Question 6 · part a · 2 marks · Working and answer

Working and answer

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15. Question 1 · part whole · 6 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

1 The nth term of an arithmetic series A is a_n

The nth term of a geometric series G is t_n

For these two series

$$a_1 = t_1 \qquad a_{10} = t_3 = 48 \qquad a_{10} = 4t_2$$

Find

(i) the common ratio of G,

(ii) the common difference of A.

(6)

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(Total for Question 1 is 6 marks)

15. Question 1 · part whole · 6 marks · Working and answer

Working and answer

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16. Question 7 · part a · 5 marks

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Further practice · Attempt part a only.

7 A geometric series has first term $(x - 3)$, second term $(x + 1)$ and third term $(4x - 2)$.

(a) Find the two possible values of x.

(5)

Given that $x < 1$,

(b) show that the series is convergent.

(2)

The sum to infinity of the series is S.

(c) Find the value of S.

(2)

The sum of the first n terms of the series is S_n

Given that $\frac{S}{S_n} = \frac{256}{255}$

(d) find the value of n.

(3)

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16. Question 7 · part a · 5 marks · Working and answer

Working and answer

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17. Question 6 · part a · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part a only.

6 The nth term of a geometric series G is U_n

The first three terms of G are given by

$$U_1 = q(4p + 1) \qquad U_2 = q(2p + 3) \qquad U_3 = q(2p - 3)$$

(a) Find the possible values of p

(5)

Given that G is convergent with sum to infinity 250

(b) find the value of q

(3)

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17. Question 6 · part a · 5 marks · Working and answer

Working and answer

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